

VocaVision: Educationally Constrained Multi-Agent Video Generation for Children’s Foreign-Language Vocabulary Learning

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Abstract

Children’s foreign-language vocabulary learning benefits from materials that bind words to memorable contexts, visual cues, spoken language, and opportunities for retrieval. Yet, common materials often fall short: word lists and static flashcards lack the vividness needed to be memorable, while generic short videos and existing cartoons are rarely designed for vocabulary acquisition and often introduce extraneous visual distractions. To bridge this gap, We present VocaVision, an educationally constrained multi-agent system that turns target English words into short picture-book-style videos for children’s vocabulary learning. Driven by multi-agent iterative optimization, VocaVision generates learning videos and retrieval exercises where memory-enhancing plots and stylistically consistent visuals seamlessly combine to provide prominent visual grounding for target words. The system design is theoretically grounded in multimedia learning, cognitive load and dual-coding principles, while technically expanding upon recent advancements in generative story-based tools. We further conducted a formative user study with eighth-grade students at Shanghai Yucai Junior High School. Their feedback suggests that VocaVision has practical value as a supportive tool for children’s English vocabulary learning. The VocaVision source code and multimedia artifacts are available at <https://github.com/Lin-Jia-Hui24/VocaVision>.

1 Introduction

Vocabulary acquisition is the cornerstone of language learning, yet traditional memorization methods often fall short of engaging modern learners. Conventional rote learning or flashcard-based approaches strip words of their natural linguistic contexts, leading to low retention rates and cognitive fatigue (Craik and Lockhart, 1972; Oxford and Scarcella, 1994). While video-assisted learning effectively enhances memory through audio-visual cues (Mayer, 1997), manually creating tailored videos for custom word lists is too time-consuming and resource-intensive. Consequently, there is an urgent need for an automated, scalable, and personalized system capable of transforming static word lists into dynamic, multi-modal learning experiences.

Recent systems can dynamically generate sentences, stories, and multimodal narratives (Leong et al., 2024; Lee et al., 2024; Arif et al., 2026), while AI-infused cartoon platforms highlight the importance of interaction and contextual expansion (Zeng and Xiong, 2026). However, existing cartoons and educational videos are rarely designed around a learner’s specific target words and often contain irrelevant visual elements that distract from learning.

To address these limitations, we introduce VocaVision, an autonomous multi-agent framework that bridges the gap between creative video generation and pedagogical utility. Designed around cognitive psychology (Sweller, 1994; 2024) and multimedia learning theories (Mayer, 1997), VocaVision orchestrates a team of specialized agents into a seamless, iterative pipeline. Given a user-defined word list, the system first initiates pedagogical context planning, where a linguistic agent grounds each word into a specific target sense, and a

Playwright Agent translates these into a narrative script, which is then dynamically refined through multi-turn collaborations with an Educator Agent. Next, in the visual synthesis phase, a Character Director Agent and a Scene Director Agent collaborate to deconstruct the approved script—the former defining character specifications and generating character references, while the latter crafts scene prompts and establishes keyframes. These visual assets are then meticulously vetted and adjusted through localized and global reviewer agents during a multi-level quality assurance process, before being finally assembled into a comprehensive learning package that delivers the narrative video, cloze-style reviews, and interactive practice questions.

The primary contributions of this work are summarized as follows:

- We formulate target-word-conditioned picture-book video generation as a vocabulary-learning format that jointly generates story context, visual anchors, narration, subtitles, and review exercises.
- We design an educationally constrained multi-agent pipeline that separates lexical grounding, learning-mode planning, story generation, educational review, visual prompting, local/global visual review, and retrieval-practice generation.
- We show how local and global visual review loops can act as educational guardrails for generated vocabulary videos, detecting target-word text errors, scene-script mismatches, and cross-scene continuity failures before final video assembly.
- We extend generated video exposure into a review loop through cloze-style videos, storybook review cards, practice questions, and feedback summaries.

2 Related Work

2.1 LLM and Multi-Agent Systems in Education

Recent advancements in Large Language Models (LLMs) have revolutionized Intelligent Tutoring Systems (ITS) and instructional design. To handle complex pedagogical tasks, researchers have increasingly shifted from single-LLM pipelines to Multi-Agent Systems (MAS). For instance, frameworks like EduPlanner (Zhang et al., 2025) utilize LLM-based multi-agent collaboration to break down and execute customized, intelligent instructional designs, while EduMAS (Li et al., 2024) leverages a specialized multi-agent framework to provide scaled educational support. Furthermore, current intelligent tutoring paradigms exploit LLM-powered multi-agent frameworks to drive goal-oriented learning (Wang et al., 2025), dynamically guiding students through structured learning paths. However, while these systems excel at textual instruction, task planning, and structural tutoring, they are primarily confined to text-based dialogues or static lesson planning, thereby lacking the capability to collaboratively translate abstract instructional goals into integrated, multimodal educational assets. To address this limitation, our framework extends the multi-agent paradigm into multimodal production for language learning. By dividing roles into specialized tasks such as narrative creation, visual choreography, and pedagogical assessment, our proposed system VocaVision autonomously transforms a raw vocabulary list into a cohesive, multimodal learning pipeline that outputs structured videos and practice questions.

2.2 Automated Storytelling and Multimodal Video Generation

Automated multimedia content generation has evolved rapidly, moving from single-model end-to-end generations to complex multi-agent orchestration across text, image, and audio. Early co-creative platforms like Kahaani (Arif et al., 2026) focused on human-AI collaborative multimodal storytelling. More recently, frameworks like StoryAgent (Hu et al., 2024) and AutoStory (Wang et al., 2024) have demonstrated the ability to generate diverse storytelling images and customized videos via multi-agent collaboration. In the educational domain, systems like MM-StoryAgent (Xu et al., 2025) generate immersive narrated storybook videos, while LASEV (Yan et al., 2026) has highlighted the superiority of LLM-based multi-agent systems for generating structured educational videos. Despite their proficiency

in creating visually appealing narratives, these systems are fundamentally entertainment-driven or general-purpose, meaning they lack pedagogical constraints and fail to map the generated animations directly to specific linguistic goals for memory retention. We close this gap by introducing target-word-conditioned generation. By grounding vocabulary in explicit senses and incorporating multi-level agent reviews, our system ensures that visual assets, script pacing, and audio narration are not only creatively coherent but also designed to reinforce the semantic meaning and cognitive retention of the target vocabulary.

2.3 Contextualized Language Learning and Retrieval Practice

Our work is firmly grounded in cognitive psychology and multimedia learning theories. Neuroscientific evidence shows that states of curiosity modulate hippocampus-dependent learning via the dopaminergic circuit, meaning that engaging narratives stimulate memory retention (Gruber et al., 2014; Prayogo et al., 2023). According to Sweller’s Cognitive Load Theory (Sweller, 2024; 1994), instructional designs can minimize extraneous cognitive load to prevent learning difficulties. Mayer’s multimedia learning work argues that combining text and visuals optimizes the dual-coding channels of the human brain (Mayer, 1997), while modern Emotional Design (Tien et al., 2018) confirms that multi-dimensional concepts and animations positively impact user affect and learning outcomes. As highlighted by recent systematic literature reviews (Shu et al., 2025), AI-generated instructional videos are becoming vital tools for realizing these theories at scale. Although the combination of contextualized storytelling and retrieval practice is theoretically optimal for language acquisition, generating such dual-phased content tailored to a student’s exact individual vocabulary weak points is labor-intensive. Consequently, our system materializes these cognitive theories into a scalable engineering reality, combining automated contextualized storytelling with instantly generated retrieval exercises to form a customizable cognitive loop for any set of vocabulary words.

3 System Overview

3.1 Task Formulation

We formulate vocabulary acquisition not as the rote memorization of isolated words, but as the generation of an immersive, multimodal educational artifact. Given a target vocabulary set, the core task of VocaVision is to synthesize a cohesive pedagogical pipeline that guides learners from contextualized exposure to active recall and transfer. Rather than treating the output as a fragmented collection of multimedia files, we unify them—contextual vocabulary videos, cloze review videos, and targeted practice questions—into an integrated learning experience (Figure 1).

Crucially, our formulation recognizes that vocabulary sets naturally vary in size, semantic density, and specific learning objectives. To accommodate these variations, we parameterize the generation task across three distinct learning modes (Figure 2): (1) Deep Single Word, which expands a seed word into a broader word family (e.g., part-of-speech variations, synonyms, and antonyms), allowing learners to acquire multiple related words simultaneously; (2) Theme Story, which weaves a small word group into a protagonist-driven narrative plot; and (3) Vocab Sprint, which utilizes brisk visual pacing for larger word sets to prioritize coverage. By formulating the task around these specific modes, VocaVision ensures that every word is presented in a cognitively optimal context rather than forced into a rigid, one-size-fits-all narrative.

3.2 Pipeline and Multi-Agent Architecture

VocaVision operates as a multi-agent orchestration pipeline designed to enforce educational constraints at every stage. The system does not ask one model to generate an entire video directly. Instead, it decomposes the learning material into a sequence of accountable decisions: lexical grounding, learning-structure planning, script generation, educational review, visual prompt design, local keyframe review, global storyboard review, media assembly, and

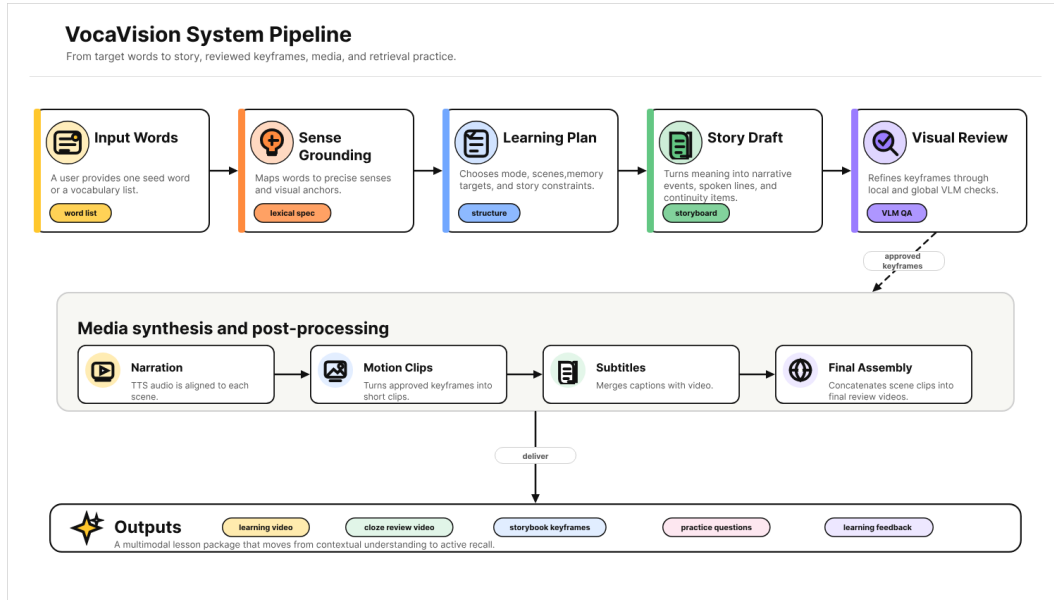


Figure 1: VocaVision pipeline. The system transforms target words into a complete learning package through lexical grounding, story planning, educational review, visual generation and review, media assembly, and retrieval-oriented exercises.

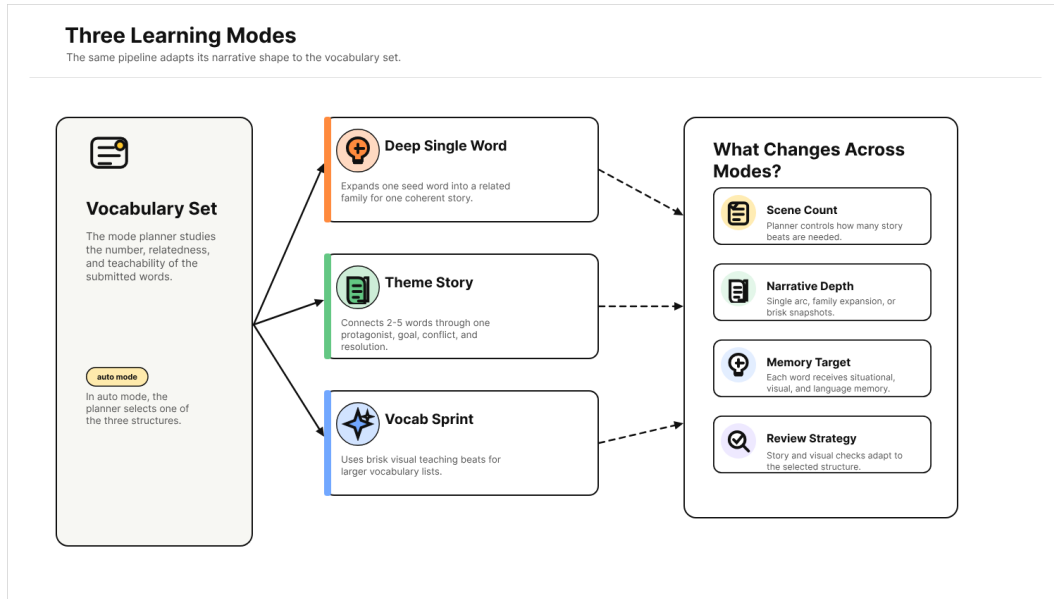


Figure 2: Three learning structures supported by VocaVision: deep single-word learning, theme-story learning, and vocabulary sprint learning.

post-video practice. Each agent receives the decisions made by previous agents as constraints and returns either an approved artifact or actionable feedback for repair.

Crucially, VocaVision orchestrates its agents not as a strict linear pipeline, but as a dynamic, multi-loop revision network. In this architecture, upstream decisions rigorously constrain downstream generation, while downstream reviewers can trigger upstream rewrites. At the

Agent	Main responsibility	Inputs and outputs
Sense Disambiguator	Select a teachable sense and visual anchors for each target word.	Takes target words and optional user hints; outputs word specifications that constrain planning, scripts, and visuals.
Learning Mode Planner	Choose the lesson structure and scene plan.	Takes word specifications and vocabulary-set size; outputs a learning plan for the Playwright Agent.
Related Word Expansion Agent	Expand a single seed word into a small related-word family.	Takes a seed word and selected sense; outputs additional teachable words for deep single-word learning.
Playwright Agent	Write scene-level narration, plot descriptions, and continuity items.	Takes the learning plan, word specifications, and educator feedback; outputs a draft script for review.
Educator Agent	Review lexical correctness, teachability, safety, and visual executability.	Takes the draft script and learning plan; outputs a pass decision or repair feedback for the Playwright Agent.
Director Agent	Translate approved scenes into character and keyframe prompts.	Takes the approved script and review constraints; outputs visual prompts revised after visual-review feedback.
Local Visual Reviewer	Check a single keyframe against its scene and target word.	Takes a generated keyframe, scene script, and word specification; returns scene-level repair feedback.
Global Visual Reviewer	Check cross-scene continuity and storyboard coherence.	Takes locally accepted keyframes; returns targeted regeneration feedback for inconsistent scenes.
Teaching Agent	Generate review activities and practice questions.	Takes the approved story, scenes, and media context; outputs cloze challenges, transfer questions, and learner feedback.

Table 1: Major agents in VocaVision and their roles in the educationally constrained generation pipeline.

foundation, the Sense Disambiguator sets the semantic and visual anchors for each word, acting as a global constraint for all subsequent steps. Guided by these anchors, the Learning Mode Planner dictates the pedagogical scaffolding that the Playwright Agent must follow to construct the narrative.

To ensure quality, the system employs two critical feedback loops. First, the Educator Agent evaluates the script, capable of rejecting drafts and routing scene-level critiques back to the Playwright. Once the script passes, the Director Agent translates it into visual prompts, initiating the second loop: the Local and Global Visual Reviewers inspect the outputs and can demand prompt adjustments, text repairs, or even full scene regenerations to maintain visual coherence. Finally, closing the loop, the Teaching Agent leverages the finalized story and media to design review activities that point learners directly back to the relevant narrative context. Figures 3 and 4 visualize these two major revision loops.

Table 1 summarizes the major agents in VocaVision and clarifies how each stage contributes to the overall educational constraint chain.

3.2.1 Script Generation and Pedagogical Planning Agents

The script-generation side of VocaVision begins with the Sense Disambiguator. Many English words are ambiguous, and a vocabulary video that depicts the wrong sense can actively

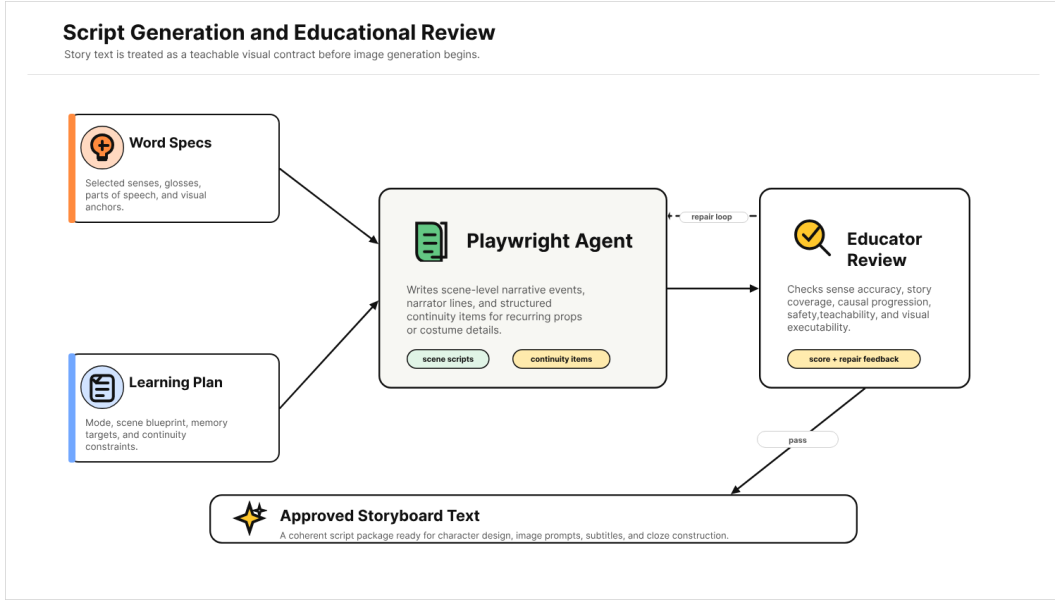


Figure 3: Script generation and pedagogical review. The Playwright Agent and Educator Agent interact through reviewable scene scripts rather than isolated example sentences.

mislead learners. The disambiguation stage therefore chooses a teachable sense, identifies plausible distractor senses, and attaches visual anchors and negative anchors that later agents must respect. These anchors serve as a bridge between lexical meaning and visual generation.

The Learning Mode Planner then transforms the word list into a lesson structure. This step is pedagogically important because different word sets call for different memory designs. A single word may benefit from expansion into a related family; a compact set of related words may benefit from a shared narrative; a larger set may need a faster sequence of memorable scenes. When the system selects the single-word deep learning structure, the Related Word Expansion Agent proposes additional classroom-friendly words connected by derivation, contrast, synonymy, or strong association.

The Playwright Agent turns the learning plan into scene-level story text. Its goal is not merely to write a pleasant story, but to create a teachable situation for each word. Each scene must support three forms of memory: situational memory, visual memory, and language memory. The Playwright Agent also records recurring visual elements such as costumes, props, and safety-relevant objects so that later visual generation can preserve narrative continuity.

The Educator Agent serves as a pedagogical gatekeeper. It reviews whether the story uses each target word in the intended sense, whether the sequence develops coherently, whether the story is safe and age-appropriate, and whether each scene can be rendered as a clear image. If the draft contains a hard educational failure, such as wrong-sense usage, unsafe framing, weak causal progression, or broken continuity, the Educator Agent returns targeted feedback to the Playwright Agent. The next draft therefore does not simply become more fluent; it becomes more teachable, more visually executable, and more consistent with the learning plan.

3.2.2 Visual Generation and Review Agents

The visual generation begins after the script has passed educational review. The Director Agent translates approved story scenes into visual prompts. It first establishes a reusable character design and then writes scene prompts that combine character identity, target-word

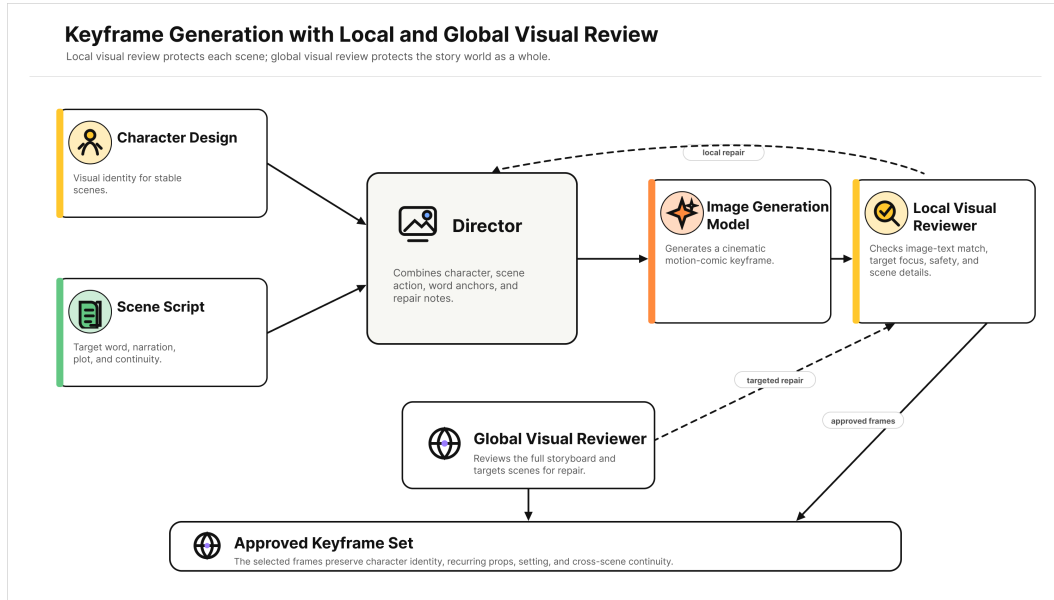


Figure 4: Keyframe generation with local and global review. Local review checks scene-level semantic alignment, while global review checks cross-scene continuity.

anchors, scene action, and continuity constraints. The Director Agent inherits constraints from both the Playwright Agent and the Educator Agent: it must preserve the intended word sense, keep recurring elements stable, and avoid visual choices that would make the lesson unsafe or ambiguous.

The Local Visual Reviewer examines each generated keyframe against the current scene script and target-word specification. It checks whether the image expresses the intended word sense, whether the main action matches the narration, whether the visual focus is clear, and whether any visible target-word text is spelled and rendered correctly. When a local mismatch is found, its feedback is passed back to the Director Agent as a repair instruction. The Director Agent then revises the prompt, optionally using the previous image or reference scenes, and the image is regenerated. This local loop protects scene-level semantic alignment.

Moving beyond individual frames, the Global Visual Reviewer assesses the visual narrative holistically. After individual keyframes have passed local review, this agent inspects the entire generated timeline to enforce cross-scene consistency—scrutinizing character identity, art style, lighting, framing, recurring props, and costume continuity. Its feedback goes beyond generic quality scores by pinpointing exactly which scenes need revisiting and which global constraints were violated. Consequently, the system can selectively regenerate problematic scenes without discarding approved assets. Ultimately, local review guarantees the pedagogical accuracy of each scene, whereas global review safeguards the holistic coherence required to anchor learner memory.

3.3 Media Assembly and Learning Exercises

Once visuals pass review, VocaVision generates narration, creates videos, and aligns subtitles with the spoken text. The final media package includes both a learning video and a cloze-style review video in which target words are masked to invite active recall.

The Teaching Agent then extends the lesson beyond video exposure. It generates scene-level cloze challenges and project-level practice questions (Figure 5). The design of these questions follows the same learning logic as the video. Cloze challenges require learners to retrieve a word from its exact visual and linguistic setting, anchoring active recall in the overarching

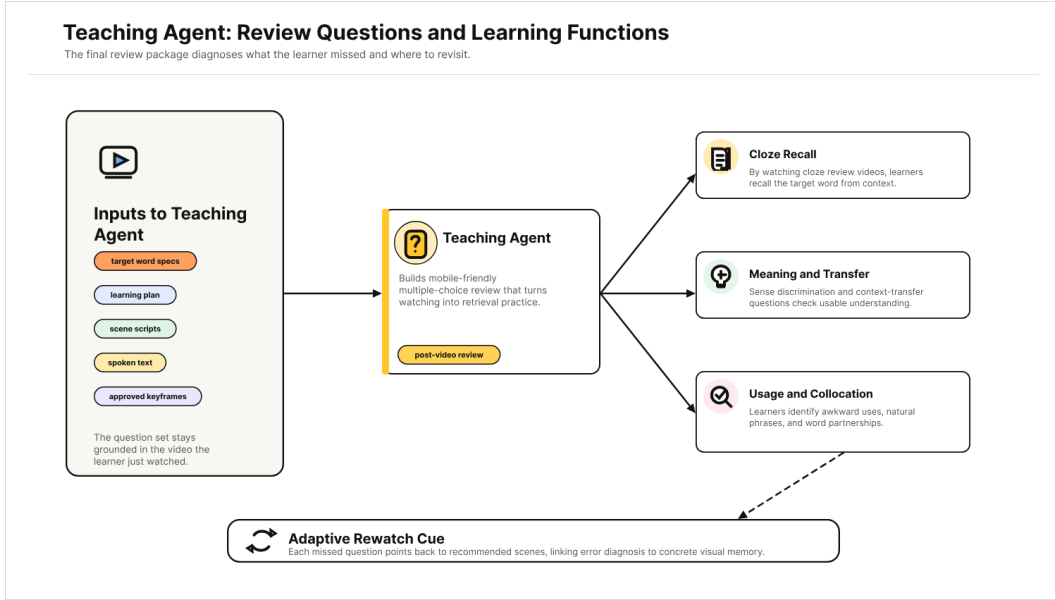


Figure 5: Teaching Agent outputs. The system produces cloze review, sense discrimination, context transfer, usage correction, and collocation-extension exercises.

narrative context. Sense-discrimination questions test whether learners can distinguish the target meaning from plausible semantic distractors. Context-transfer questions ask whether learners can recognize the word in a new situation, moving beyond memorization of the original sentence. Usage-correction questions expose awkward or incorrect uses, encouraging learners to notice grammatical and semantic constraints. Collocation-extension questions introduce natural word partnerships, helping learners move from isolated word recognition toward usable language knowledge. Together, these question types turn the generated video into a complete learning cycle: watch, remember, retrieve, diagnose, and revisit.

Figure 6 shows how these outputs are presented as a learner-facing loop rather than as disconnected media files. The generated video, picture-book review, cloze review, practice questions, feedback, and scene revisit steps are designed to move learners from exposure to active recall and correction.

4 Case Studies and User Study

4.1 Case Study: Script Review and Pedagogical Enrichment

We first examine a script-generation case for the care word family, which encompasses the specific target words care, “caring,” careful, and careless.” Figure 7 presents the concrete outputs used in the comparison. A baseline script generated by direct LLM prompting without the full VocaVision review process (Appendix B.5) produced a sequence of loosely connected events: Lily feeds a fish, helps her brother with homework, walks carefully on ice, helps Tommy after a fall, and later shares an umbrella. Although each sentence uses a target word, the sequence functions more like separate examples than a coherent learning story. It offers limited causal progression, weak recurring visual memory, and little reason for the learner to connect the words as a family.

In the first VocaVision draft, the Playwright Agent reframed the word family around a unified narrative: classmates taking care of a pet rabbit. This immediately improved story coherence, as the target words became anchored to a single setting, a specific group of characters, and a focused, care-related objective. However, the Educator Agent rejected the draft with a score of 7.8. Its feedback identified weak causal links between adjacent scenes,

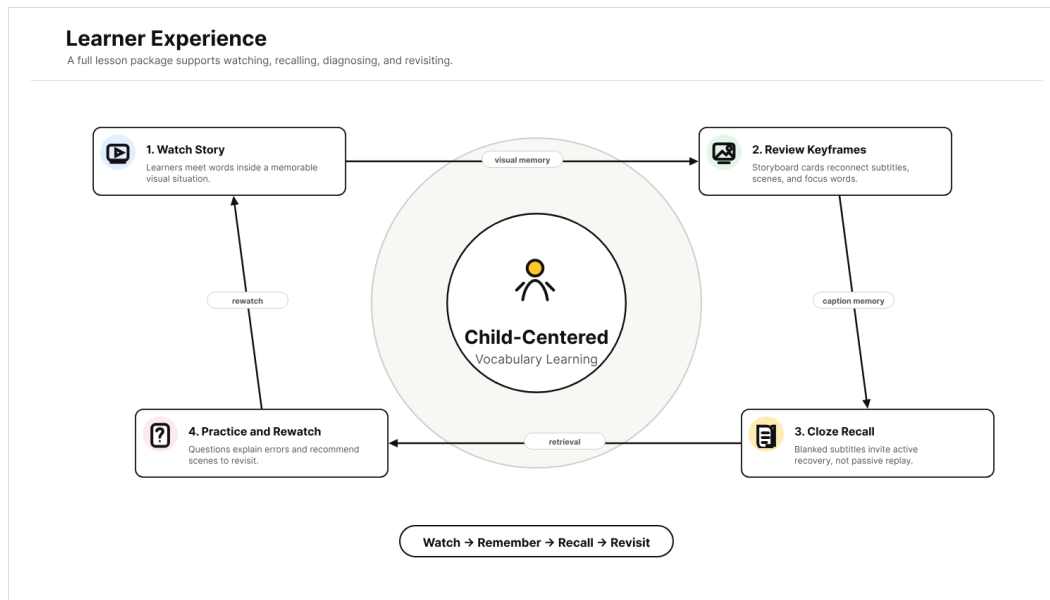


Figure 6: Learner experience loop. The generated video, storybook review, cloze review, question feedback, and scene revisit flow are designed as a continuous learning cycle.

Script Review Case Study: The Care Word Family

From loose word examples to a coherent, reviewable learning story.

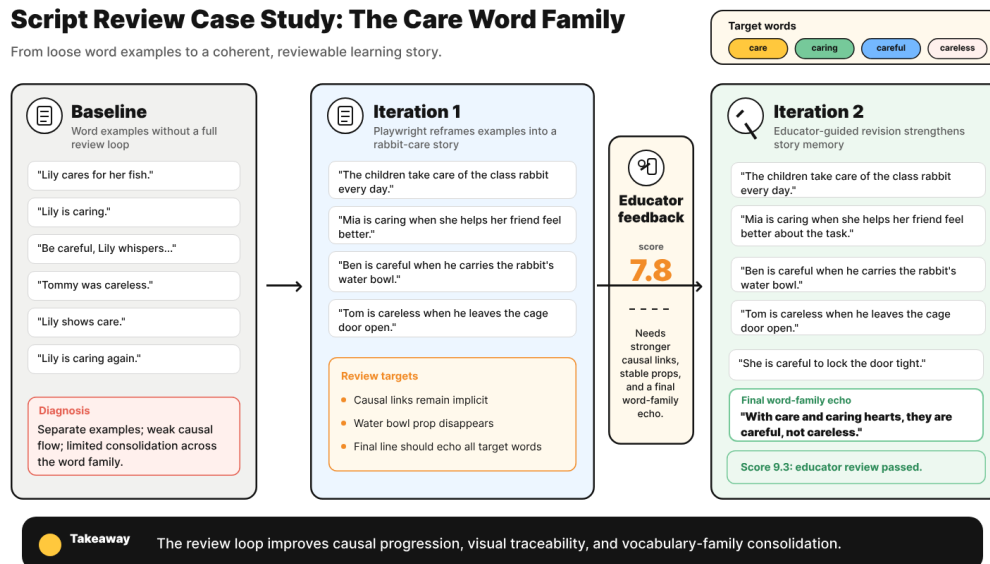


Figure 7: Script review case study for the care word family, illustrating how educator feedback iteratively improves causal progression, visual traceability, and word-family consolidation.

incomplete tracking of the water bowl prop, and an insufficient final review of the full word set.

After revision, the second iteration reached a score of 9.3. The story preserved the class-pet premise while strengthening causal progression and visual continuity. The final narrator line explicitly connected the whole word family: "With care and caring hearts, they are

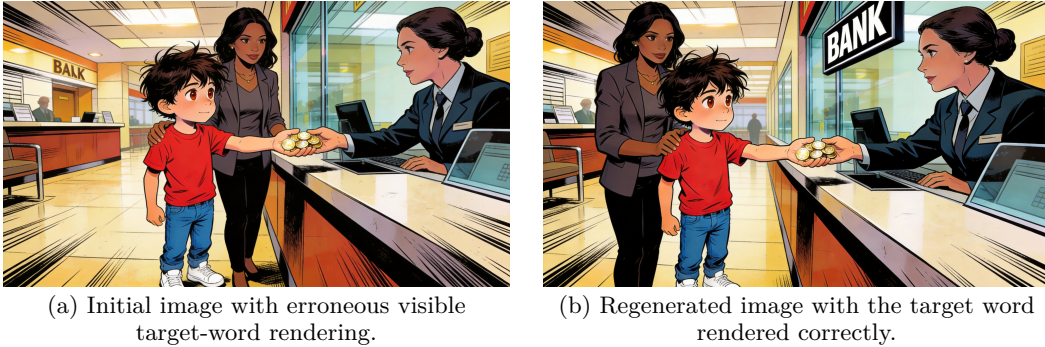


Figure 8: Local visual review case for correcting visible target-word spelling.

careful, not careless.” This case illustrates why the Playwright-Educator interaction is not redundant: the Playwright Agent provides narrative imagination, while the Educator Agent transforms that narrative into a robust learning object by enforcing causal continuity, visual traceability, and vocabulary-family consolidation.

This case can be read through the same pedagogical dimensions used in our post-study learner questionnaire. Cognitive load is reduced because the revised story no longer jumps among unrelated events. Multimedia readiness is enhanced because elements like the water bowl, the hutch, the latch, and the rabbit create stable visual anchors. Curiosity and engagement are fostered by a compelling narrative arc that progresses from a routine care-taking responsibility, to a careless mistake, and finally to a successful resolution. Perceived learning value is strengthened because the final line explicitly contrasts careful and careless while bringing care and caring back into the same cohesive memory frame.

4.2 Case Study: Correcting Target-Word Rendering

We use a visual review case to illustrate why educational guardrails are necessary. In an initial keyframe generated for the target word “bank,” the image model rendered the visible word as “BALK” rather than “BANK.” Figure 8 shows the initial error and the repaired keyframe. The Local Visual Reviewer identified this as a severe error because the target word appeared visibly in the image but was misspelled. In a vocabulary-learning video, this failure is not a minor aesthetic defect: it may directly interfere with the learner’s acquisition of the word form. The system therefore rejected the keyframe and regenerated the scene. The repaired image displayed the target word prominently and correctly.

4.3 Case Study: Global Review for Cross-Scene Character Consistency

The Global Visual Reviewer is useful when each individual keyframe may look plausible in isolation but the full sequence fails as a coherent story. In one generated storyboard, the first scene depicted a teacher and student whose hair color and skin tone were visibly inconsistent with the characters in later scenes. The Global Visual Reviewer detected this cross-scene inconsistency and targeted the first scene for regeneration. Figure 9 shows the targeted repair process. After the first repair, the teacher’s hair color became more consistent with the rest of the sequence, but the student’s skin tone still did not fully match. After a second repair, both the teacher’s hair color and the student’s skin tone were aligned with the remaining scenes, producing the reviewed sequence in Figure 10. This case shows why global review must be separated from local review: a local reviewer can decide whether a single frame matches its scene, but only a global reviewer can detect whether recurring characters remain visually stable across the whole story.



Figure 9: Targeted global repair for the inconsistent first scene.



Figure 10: Reviewed storyboard after global consistency repair.

4.4 Formative User Study

4.4.1 Research Goals and Hypotheses

To evaluate the pedagogical efficacy of our proposed framework, this formative study focuses on learner-centric experience and learning outcomes. We formulate our evaluation around the following two Research Questions (RQs):

- RQ1 (Learner Experience): How do eighth-grade students perceive the system in terms of cognitive load, multimedia learning experience, curiosity, engagement, and interactive learning support?
- RQ2 (Learning Outcomes): Does the multi-agent generated content effectively support English vocabulary acquisition, as reflected in both learner self-reports and immediate exercise performance?

Based on these questions, we establish the following hypotheses to guide our analysis:

- H_1 : The multi-agent generated animated scripts and videos will stimulate high learner curiosity and engagement while maintaining a manageable, low cognitive load.
- H_2 : Learners will perceive a positive effect on their English vocabulary retention and benefit from the integrated interactive learning exercises.

4.4.2 Participants

Forty eighth-grade students were recruited from Shanghai Yucai Junior High School. Rather than conducting a controlled, synchronous experiment inside a physical classroom, we deployed the system as a web-based platform. This allowed students to access the system and complete sessions remotely, asynchronously, and at their own pace, fostering a naturalistic and self-directed learning environment.

The student group ($N = 33$) comprised eighth-grade language learners representing a diverse distribution of prior English proficiencies and learning habits typical of the local curriculum. All participating students possessed the necessary digital literacy to independently navigate web-based applications on their personal devices (e.g., laptops or tablets) at home.

Prior to the study, informed consent was secured from the participating teachers and the legal guardians of all participating students.

4.4.3 Experimental Stimuli

Given that end-to-end multi-agent narration synthesis and video rendering require a non-trivial amount of computational time, deploying the pipeline entirely in real-time would severely degrade the interactive user experience during the evaluation. To mitigate this technical latency and foster seamless, self-paced engagement, the learning materials were pre-generated based on the official target vocabulary lists from the eighth-grade curriculum.

Student participants were provided with these curated packets, from which they could autonomously select their desired word sets for study. All students interacted with the finalized multimedia content generated by the complete multi-agent system, including the story video, picture-book review, cloze review, practice exercises, and post-study questionnaire. Because participation was voluntary and asynchronous, our analyzable sample comprised 20 completed post-study questionnaires and 26 learning sessions with logged exercise performance.

4.4.4 Evaluation Metrics

To collect empirical evidence regarding the formulated research questions and hypotheses, our evaluation framework utilizes a mixed-methods approach comprising subjective psychometric instruments and objective learning performance telemetry.

Learner Subjective Metrics (Student Questionnaire) To evaluate the end-to-end user experience of the full system from the student perspective, we designed a 14-item post-study questionnaire leveraging 5-point Likert scales (ranging from 1: “Strongly Disagree” to 5: “Strongly Agree”). Grounded in cognitive psychology and educational technology theories, these items are mathematically aggregated into five distinct constructs:

- **Cognitive Load Management (CL, 3 items):** Grounded in Cognitive Load Theory (CLT), this metric operationalizes spatial-temporal contiguity and redundancy effects to assess whether the synchronized multi-agent video effectively reduces extraneous mental effort and integration burdens.
- **Multimedia Learning Experience (ML, 3 items):** Based on Mayer’s Cognitive Theory of Multimedia Learning (CTML), this construct evaluates dual-channel processing efficiency and multimedia effects derived from the dual-modality output (the synergistic coherence of visuals, voiceovers, and subtitles).
- **Curiosity and Engagement (CD, 3 items):** This measures the extent to which unexpected narrative arcs or integrated inquiry cues stimulate state curiosity and sustain learner attention throughout the session.
- **Perceived Learning Effectiveness (LE, 3 items):** This serves as a subjective proxy indicator for immediate vocabulary retention, delayed memory durability, and potential cognitive transfer of target word knowledge into novel sentence contexts.
- **Interactive Learning Support (IS, 2 items):** Rooted in self-efficacy and emotional engagement theories, this quantifies the perceived pedagogical value of the active retrieval practice and immediate feedback loops provided during the post-video exercises.

Learner Objective Metrics (Learning and Exercise Performance) Beyond subjective self-reports, the web-based platform continuously captured behavioral telemetry data to objectively evaluate vocabulary acquisition. Upon completing each video-based learning block, students were required to complete system-generated interactive exercises. We define three primary objective metrics:

- **Overall Exercise Accuracy (Acc_{all}):** The aggregated correct response rate across all question types within a single learning session, serving as a holistic indicator of immediate learning outcomes.
- **Cloze Question Accuracy (Acc_{cloze}):** The percentage of correct inputs in contextual cloze recall tasks, which specifically measures the learner’s capability to actively

retrieve and recall target vocabulary within the recently viewed narrative scenes and immediate video-based review cues.

- Practice Question Accuracy ($Acc_{practice}$): The success rate in comprehension and application exercises, which operationalizes the student’s semantic understanding and initial cognitive transfer capability by evaluating vocabulary usage entirely independent of the original video scripts or sentence-by-sentence retelling.

To evaluate the framework’s robustness across diverse learner profiles, these objective performance metrics are structured to be stratified by the participants’ self-reported English proficiency level and prior word familiarity.

4.4.5 Results

We analyze the empirical data from the user study to evaluate the system’s pedagogical efficacy, encompassing subjective survey metrics and objective exercise performance.

Subjective Perception and User Experience The subjective user experience was quantified via the post-study questionnaire ($n_{survey} = 20$). Overall, the multi-agent system achieved high user acceptance, yielding an aggregate mean score of $\mu = 4.7321$ (Median = 5.0) across all 14 items. Table 2 details the descriptive statistics across the five pedagogical constructs.

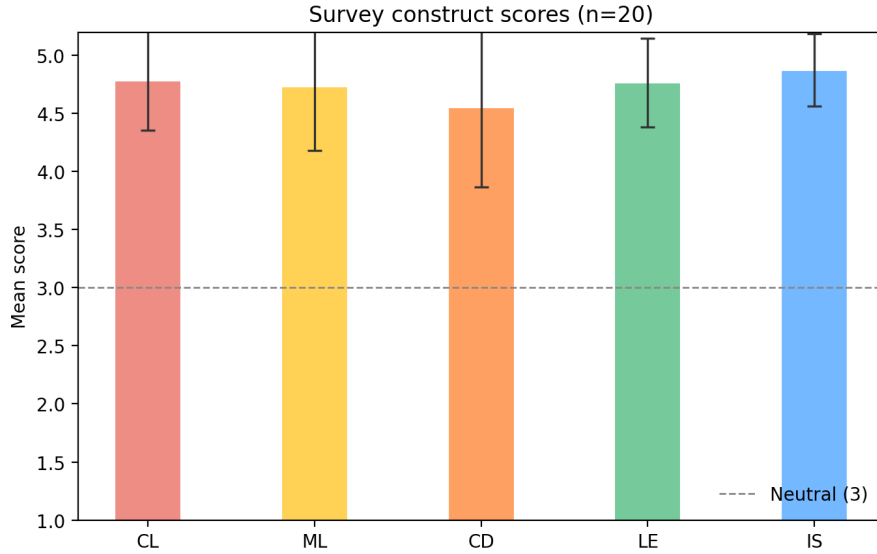


Figure 11: Mean scores across the five survey constructs. Error bars denote the standard deviation (σ), and the dashed line indicates the scale’s neutral baseline (3.0).

Construct / Dimension	Items	Mean (μ)	SD (σ)	Median
Cognitive Load Management (CL)	3	4.7833	0.4253	5.0
Multimedia Learning Experience (ML)	3	4.7333	0.5538	5.0
Curiosity and Engagement (CD)	3	4.5500	0.6854	5.0
Perceived Learning Effectiveness (LE)	3	4.7667	0.3815	5.0
Interactive Learning Support (IS)	2	4.8750	0.3112	5.0

The score distribution highlights key design strengths:

- Instructional Scaffolding: Interactive Learning Support (IS) scored the highest ($\mu = 4.875$), indicating that immediate feedback reinforcement effectively enhanced learner confidence.

- Low Cognitive Overhead: The high score in Cognitive Load Management (CL) ($\mu = 4.783$) confirms that multi-agent synchronization minimized extraneous load and prevented split-attention effects.

Objective Learning and Transfer Performance The system systematically logged student performance data across 26 completed learning sessions to objectively evaluate vocabulary acquisition outcomes.

Overall, students achieved a robust comprehensive accuracy rate of $Acc_{all} = 0.7847$. A deeper inspection into specific exercise typologies reveals a distinct hierarchical proficiency pattern across the cohort:

- Contextual Recall Proficiency: Students demonstrated exceptional accuracy in Cloze Recall tasks ($Acc_{cloze} = 0.9500$). This near-perfect performance proves that the generative narrative contexts and synchronized multi-modal animations established exceptionally strong semantic anchoring in immediate memory, facilitating highly efficient, cued lexical retrieval.
- Cognitive Transfer Challenges: Conversely, the application tasks yielded a more moderated success rate ($Acc_{practice} = 0.6000$). Because these practice items required learners to deploy the target vocabulary in entirely novel sentence structures independent of the original video storyline, this decline accurately reflects the expected cognitive barrier of moving from baseline lexical recognition to deep, productive language transfer.

Stratified Analysis by Learner Profiles To evaluate system adaptability, sessions were stratified by self-reported English proficiency and prior word familiarity (Table 3).

Table 3: Stratified Performance Tracking by Learner Profiles

English Level	Word Familiarity	Sessions (n)	Acc_{all}	Acc_{cloze}	$Acc_{practice}$
Confident	Known	15	0.8884	0.9867	0.7026
Confident	New	1	0.4000	1.0000	0.0000
Middle	Known	4	0.7000	0.8500	0.6667
Middle	Mixed	6	0.6463	0.9167	0.4445

The stratified data reveals two major insights:

- Scaffolding Lower-Ability Learners: Critically, “Middle” proficiency students tackling “Mixed” vocabularies containing unfamiliar words ($n = 6$) maintained a high cloze accuracy ($Acc_{cloze} = 0.9167$). This demonstrates that the multi-modal narrative context serves as an effective cognitive scaffold for struggling learners.
- The Application Bottleneck: The drop in practice accuracy for the “Middle-Mixed” cohort ($Acc_{practice} = 0.4445$) highlights the steep curve of cognitive transfer. Similarly, a single outlier case (Confident-New, $n = 1$) achieved perfect cloze recall ($Acc_{cloze} = 1.0000$) but failed the semantic transfer task ($Acc_{practice} = 0.0000$). This contrast underscores that while multi-agent video artifacts excel at cued memory retrieval, scaling deeper semantic transfer for entirely unfamiliar words requires more iterative, cross-contextual application.

5 Discussion

5.1 From Generated Content to Pedagogical Artifacts: The Need for Auditability

VocaVision highlights a critical distinction between generated content and generated learning materials. Current generative models can easily produce videos that are fluent, colorful, and highly engaging, yet completely fail a pedagogical objective. An LLM or video model might blindly use the wrong word sense, omit the target word altogether, clutter the scene

with irrelevant distractors, or introduce inconsistent props that break the learner’s memory cues.

This limitation points to a broader design implication for the intersection of AI and education: generative AI in educational contexts should not operate as a black box that merely outputs final media files; instead, it must yield interpretable, verifiable, and auditable learning materials. VocaVision operationalizes this principle by distributing educational responsibility across distinct agents and transparent review loops. Because our architecture generates explicit intermediate representations—such as sense disambiguation constraints, numerical pedagogical scores, and targeted visual repair instructions—every generative step becomes fully transparent. Crucially, this traceability empowers human teachers to intervene, review, and manually edit the content at any stage. This shift from a black-box “single-shot generation” to an “editable pedagogical pipeline” ensures that human educators, rather than unpredictable AI algorithms, ultimately dictate the final learning experience.

5.2 Why Custom Video Rather Than Existing Cartoons?

Existing cartoons can be engaging, and prior work such as AnimAlte shows that they can be used as vocabulary-learning resources (Zeng and Xiong, 2026). However, a pre-existing cartoon was not designed for a specific target word set. Its plot, props, dialogue, pacing, and visual emphasis may serve entertainment or narrative goals rather than the learner’s vocabulary goal.

VocaVision instead creates video around target words. This design has theoretical significance. From a cognitive load perspective, customized scenes can reduce extraneous information because every major visual element can be selected to support the target word. From a multimedia learning perspective, narration, subtitles, and visuals can be aligned around the same semantic focus rather than competing for attention. From a dual-coding perspective, each word can be paired with a diagnostic visual situation and a repeatable language cue. From a vocabulary-learning perspective, the system can connect exposure, contextual understanding, active recall, and transfer practice in one continuous learning flow.

5.3 Limitations and Future Directions

VocaVision currently focuses on English vocabulary learning for beginning and intermediate foreign-language learners. The system does not yet maintain a long-term learner model, so it cannot fully adapt to a child’s changing proficiency, memory history, preferences, or error patterns across many learning sessions. Future work should incorporate learner profiles, spaced review schedules, and longitudinal tracking so that generated videos and exercises can respond to individual learning trajectories.

The current system also relies on automated review agents. Although these agents reduce many risks, they cannot replace human educational judgment, especially when materials are used with children. Future deployments should support teacher review, editable outputs, and classroom-level customization. In addition, the present study is formative; stronger evidence of learning effectiveness will require controlled experiments with pre-tests, post-tests, delayed retention tests, and comparisons against established vocabulary-learning materials.

Another limitation is language specificity. VocaVision currently targets English vocabulary learning, but different languages have various morphological systems, usage patterns, cultural contexts, and learning difficulties. Future work can explore how the system should be adapted for other languages, including how to design visual anchors, story structures, and review questions that fit language-specific learning patterns.

Finally, VocaVision generates a constrained motion-comic style rather than fully expressive animation. This choice improves controllability and visual consistency, but it may limit emotional nuance and scene complexity. Future work can explore richer animation while preserving the educational guardrails that make target-word learning reliable.

5.4 Ethics and Safety

Because VocaVision is designed for children’s learning, ethical and safety considerations are central to its architecture rather than optional afterthoughts. In our research process, we strictly minimized personal data collection, protected participant privacy, and enforced rigorous consent and anonymization procedures during the user study. Within the system itself, the multi-agent review loops actively screen generated stories and visuals to prevent age-inappropriate content, stereotypes, unsafe behaviors and misleading language.

Furthermore, we explicitly position VocaVision as an assistive authoring tool rather than a replacement for educators or caregivers. To support essential human oversight, all generated materials remain fully editable and reviewable before classroom use. While the system supports targeted vocabulary generation, its design intentionally avoids inferring sensitive user attributes or overfitting content to private learner data.

6 Conclusion

We presented VocaVision, an educationally constrained multi-agent system for generating picture-book-style vocabulary learning videos. The system grounds target words in confirmed senses, constructs story contexts, reviews educational and visual quality, assembles synchronized narration and subtitles, and extends exposure into cloze and practice exercises. Ultimately, VocaVision illustrates how generative media can become more practical for children’s language learning when video generation is treated as a constrained learning-design process. In future work, we hope to strengthen learner modeling, evaluate long-term vocabulary retention, support language-specific teaching designs, and develop more flexible teacher-guided authoring workflows.

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A User Study Questionnaire

This appendix provides the English version of the post-study questionnaire used in the formative learner evaluation. The instrument captures subjective perceptions of cognitive load, multimedia learning quality, curiosity and engagement, perceived learning effectiveness, and interactive learning support after students complete a VocaVision learning session.

A.1 Design Rationale

The questionnaire is organized around five learner-experience constructs:

Cognitive Load Management. This dimension examines whether VocaVision reduces unnecessary effort during vocabulary learning. It is motivated by cognitive load theory and multimedia learning principles, especially the need to align narration, visuals, and subtitles while avoiding redundant or distracting information.

Multimedia Learning Experience. This dimension examines whether visual, auditory, and textual channels support each other. It is motivated by the cognitive theory of multimedia learning and dual-channel processing.

Curiosity and Engagement. This dimension examines whether the generated story maintains attention and encourages learners to continue watching. It is motivated by curiosity-driven learning and prediction-based engagement.

Perceived Learning Effect. This dimension examines learners’ subjective sense of vocabulary acquisition, including immediate recall, expected delayed retention, and transfer to new contexts.

Interactive Learning Support. This dimension examines whether post-video questions and feedback help learners consolidate vocabulary knowledge through active recall, feedback, and self-efficacy.

A.2 Student Questionnaire Items

Participants were instructed to rate each statement based on their experience of learning English words with the video they had just watched. The response scale was: 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree.

Optional open responses collected the learner’s most memorable target word and free-text comments about the session.

B Agent Prompt Specifications

This appendix summarizes the prompt specifications used by VocaVision’s multi-agent pipeline. The purpose of including these specifications is to make the system design auditable: each agent is assigned a bounded educational responsibility, a set of constraints, and a structured output contract. The implementation uses longer prompt templates, but the following descriptions preserve the core role definitions, review criteria, and agent interactions needed to reproduce the design.

Item	Dimension	Statement
CL-1	Cognitive load management	The pictures and narration in the video worked well together, so I could understand the word meaning without watching it repeatedly.
CL-2	Cognitive load management	The video looked clean and did not include extra pictures or text that distracted me.
CL-3	Cognitive load management	Learning words through this video felt easier than learning from a book or only listening to an explanation.
ML-1	Multimedia learning experience	The animation helped me understand the word meaning, instead of relying only on spoken explanation.
ML-2	Multimedia learning experience	The pronunciation of the word and the matching picture appeared at the right time, so I was not confused.
ML-3	Multimedia learning experience	The animation and sound worked better together than only looking at pictures or only listening to explanation.
CD-1	Curiosity and engagement	While watching the video, I was curious about what word or picture would appear next.
CD-2	Curiosity and engagement	Small surprises or questions in the video made me want to keep watching.
CD-3	Curiosity and engagement	During the learning activity, I felt focused and did not get distracted easily.
LE-1	Perceived learning effect	After watching the video, I could clearly remember the meaning of the word.
LE-2	Perceived learning effect	I think I would still remember the word after a few days.
LE-3	Perceived learning effect	I could recognize and use the word correctly in a new sentence or situation.
IS-1	Interactive learning support	The questions during or after the video helped me strengthen my memory of the word.
IS-2	Interactive learning support	When I received feedback on my answer, I felt more confident to continue learning.

Table A.1: Student questionnaire items ($n = 14$ Likert items).

B.1 Sense Disambiguator Agent

Role. Resolve the intended teachable sense of each target word before story generation begins.

Core instructions. Infer the most teachable sense for a short educational story; provide multiple candidate senses when a word is ambiguous; avoid merging multiple senses into one lesson; provide visual anchors and negative anchors; and mark whether user confirmation is recommended.

Output contract. A structured list of word specifications, including the selected sense, part of speech, glosses, candidate senses, visual anchors, negative anchors, confidence, and confirmation status.

B.2 Learning Mode Planner

Role. Choose the pedagogical structure that best fits the submitted vocabulary set.

Core instructions. Prefer deep single-word learning when one seed word can be expanded into a teachable word family; prefer a theme story when two to five words can support one coherent narrative world; prefer a sprint structure when the word list is larger and needs fast, memorable coverage; recommend a scene count and explain the rationale.

Output contract. A learning plan containing the selected structure, confidence, scene count, story setting, protagonist, goal, conflict, resolution condition, memory targets, and scene blueprint.

B.3 Related Word Expansion Agent

Role. Expand one seed word into a small, classroom-friendly related-word family when deep single-word learning is selected.

Core instructions. Return four additional teachable words; prefer derivatives, synonyms, antonyms, contrasts, or strong semantic associations; avoid rare, unsafe, or visually difficult words; and keep the set diverse enough to support a meaningful story.

B.4 Playwright Agent

Role. Write the vocabulary-learning story as a sequence of teachable scenes.

Core instructions. Follow the learning plan and scene blueprint; use each target word in its selected sense; make each scene support situational memory, visual memory, and language memory; maintain causal progression; track recurring characters, props, clothing, and safety-relevant objects; avoid unsafe imitation cues and dry dictionary-like narration; and repair scenes explicitly when Educator feedback is provided.

Output contract. A structured list of scenes. Each scene includes the focus word, visual plot description, spoken narration or dialogue, and continuity items for recurring visual elements.

B.5 Direct LLM Baseline Prompt (Case Study Comparison)

The script case study in Section 4.1 compares VocaVision drafts against a direct LLM baseline: a single-turn prompt that bypasses the multi-agent pipeline and asks one language model to return a JSON script directly. This baseline corresponds to prompt version `script_ablation_baseline_v1` in our implementation.

System prompt.

Return valid JSON only.

User prompt template. Placeholders are filled at runtime from the target word list, learning mode, scene count, and narration word limit.

Write a {learning mode} script for the target word(s): {target words}.
The script will be used to generate an English learning video for 12-year-old students.
Write {scene count} scene(s). Total narration must not exceed {max words} English words.
All text must be in English.

Return JSON:

```
{
  "scenes": [
    {
      "target_word_in_scene": "word",
      "script": {
        "plot_description": "visual story beat",
        "voiceover_and_dialogue": "spoken English narration"
      }
    }
  ]
}
```

Unlike the Playwright-Educator loop, this baseline does not perform sense grounding, learning-mode planning, pedagogical review, or continuity repair. It therefore serves as a lower-bound reference for script coherence in the case study rather than as part of the deployed learner-facing system.

B.6 Educator Agent

Role. Review whether a script is suitable as a vocabulary-learning object.

Core review criteria. Sense accuracy, part-of-speech correctness, coverage of all target words, causal progression, story coherence, memory quality, transferability, child safety, age appropriateness, visual executability, and continuity readiness.

Output contract. A structured review with pass status, score, feedback, strengths, improvement focus, blocking issues, and scene-level repair suggestions.

B.7 Director Agent

Role. Translate approved story scenes into visual prompts for image generation.

Core instructions. Produce a reusable character design and scene-level keyframe prompts; preserve selected word senses and visual anchors; respect continuity constraints for clothing, props, safety gear, and recurring objects; avoid lip-sync-heavy descriptions; emphasize cinematic motion-comic composition, body language, props, and readability; and revise prompts when visual feedback is provided.

B.8 Local Visual Reviewer Agent

Role. Judge whether a generated keyframe correctly supports one scene.

Core review criteria. Image-script alignment, visibility and diagnosticity of the target-word meaning, correct use of visual anchors, avoidance of negative anchors, safety, age appropriateness, absence of distracting visual elements, and correct spelling and legibility if the target word appears as visible text.

Interaction. If the keyframe fails local review, feedback is sent to the Director Agent. The Director Agent revises the prompt, and the image is regenerated.

B.9 Global Visual Reviewer Agent

Role. Judge whether the approved keyframes work together as one coherent storyboard.

Core review criteria. Character identity consistency, art style coherence, lighting and color coherence, recurring prop and costume continuity, cross-scene visual logic, target-word readability, and absence of unsafe or pedagogically misleading visual drift.

Interaction. The Global Visual Reviewer identifies only the scenes that need repair. Its feedback is sent to the Director Agent and, when needed, back to scene wording repair.

B.10 Teaching Agent

Role. Convert the approved video lesson into retrieval-oriented review activities.

Core instructions. Prefer mobile-friendly multiple-choice questions; generate one cloze challenge per scene; keep cloze prompts close to the original narration when possible; generate practice questions that test sense discrimination, context transfer, usage correction, and collocation extension; provide plausible distractors; and recommend scenes for learners to revisit when they miss a question.

B.11 Prompt Design Principles

Across agents, VocaVision follows four prompt-design principles: educational specificity, structured output, repairability, and child-centered safety. Prompts describe what counts as a learning failure, not only what counts as a generation failure; agents return machine-readable decisions; review agents provide targeted feedback; and all story, image, and exercise prompts require age-appropriate, safe, and classroom-suitable content.

C Team Members and Contributions

This project was a highly collaborative effort. The specific contributions and division of labor for each team member are detailed below:

- Jiahui Lin (Student ID: 24307140111)
 - System & Engineering: Led the core system design of VocaVision, developed the backend framework, implemented the frontend user interface, and prepared the final presentation slides.
 - Paper Writing: Primary author for the System Overview, Case Study and User Study, and Discussion sections.
- Lingyue Ren (Student ID: 24307140115)
 - Research & Evaluation: Conducted the literature review, managed website deployment, designed the user study questionnaire, analyzed user experiment data, and co-prepared the presentation slides.
 - Paper Writing: Primary author for the Introduction, Related Work, and Case Study and User Study sections.